

SCORING, SUBSTITUTION MATRICES AND SCORE AGGREGATION FOR TCR ANTIGEN-SPECIFICITY SEARCH

The `vdjmatch` authors*

vdjmatch technical appendix — draft, regenerated from `bench/gen_vdjam.py`, `bench/loo_vdjam.py`,
`bench/gen_appendix_assets.py`

Abstract

`vdjmatch` annotates T-cell receptor (TCR) clonotypes by antigen specificity through fuzzy CDR3 search against a labelled database (VDJdb), built on the `seqtree` search core. This note documents the *scoring* layer: how CDR3 similarity is scored, the data-driven TCR substitution matrix (VDJAM) and why it is region-structured, and how per-hit scores are aggregated into a control-calibrated significance. CDR3 loops resist the assumptions behind BLOSUM/PAM — they cannot be multiply aligned across clonotypes of different length, and germline-encoded flanks (CASS...) are so over-conserved that a naive matrix conflates germline bias with selection. We give a position-debiased log-odds estimator that isolates the substitution signal, show empirically that this signal lives in the non-template (NDN) core while the V/J flanks are germline-determined, and demonstrate the consequences on worked examples (GIL, NLV) using the alignments and CIGAR strings produced by `seqtree`. The E-value theory used for significance is derived in the companion `seqtree` appendix; here we summarize it and give the paired-chain extension.

1. THE SEARCH-AND-SCORE FRAMEWORK

Given a query CDR3 q and a labelled reference set D (VDJdb), `vdjmatch` retrieves the neighbours of q within a fixed scope $\theta = (s, i, d, t)$ (max substitutions, insertions, deletions, total edits) using the `seqtree` engine, which defines a ball $B_\theta(q) = \{x : \text{edit}(q, x) \leq \theta\}$ and returns, per hit, the edit decomposition and a matrix-weighted alignment. The matrix enters as a *ranking* score inside the ball; the ball itself (an edit budget) bounds the candidate set. Each hit carries its antigen labels, an extended CIGAR (`=/X/I/D`), and an alignment score. Significance comes from comparing the neighbour count to a matched background control (§3).

2. SUBSTITUTION SCORING AND THE VDJAM MATRIX

HOW BLOSUM IS BUILT, AND WHY IT DOES NOT TRANSFER.. BLOSUM is a log-odds matrix estimated from the BLOCKS database of *ungapped multiple-alignment blocks* of conserved protein regions. Its recipe is: (i) take many trusted blocks — columns of homologous, gaplessly aligned

*Supplementary technical note for the `vdjmatch` software; intended as supplementary material to the main manuscript.

residues; (ii) cluster sequences within a block above a percent-identity threshold and down-weight within-cluster pairs so abundant near-duplicates do not dominate; (iii) count, over all blocks, how often residues a and b co-occur in the *same alignment column*, q_{ab} ; (iv) score $s(a, b) = \frac{1}{\lambda} \log(q_{ab}/p_a p_b)$ against the *global* background frequencies p_a . The signal is “which residues substitute for one another within conserved, column-homologous positions,” and the background is position-independent.

CDR3 loops break every step. (i) There is no trusted multiple alignment: same-specificity CDR3s differ in length and have no column homology, so “alignment columns” do not exist. (ii) The germline-encoded ends are over-conserved — essentially every human TRB CDR3 begins CASS and ends in a short J motif — so a *global* background p_a is wrong: an alanine in the N-terminal CASS is germline-fixed and carries no specificity signal, while an alanine in the centre does, yet BLOSUM would score them identically. A single global matrix estimated over all positions thus conflates germline conservation with antigen-driven selection.

WHAT VDJAM CHANGES (POINT BY POINT).. VDJAM keeps the log-odds idea but replaces the two broken ingredients. (a) *Observation unit*: instead of columns of a multiple alignment, the unit is a *same-antigen single-substitution pair* — two CDR3s annotated to the same epitope, equal length, differing at exactly one position; this is the only “aligned residue pair” obtainable without a multiple alignment, and (like a BLOCKS column pair) it records two residues that are interchangeable without changing function (here, specificity). (b) *Background*: instead of a global p_a , a *position-specific* background $\text{bg}(\cdot | L, p)$ (the residue frequency at that length and position) is used, so over-conserved columns are normalized away analytically rather than masked by hand. (c) *Redundancy*: BLOSUM’s percent-identity clustering is replaced — partly by the position-specific background (germline-shared residues are “expected” and contribute little), and prospectively by the control-calibrated significance layer (§3) that already discounts convergent public clones; per-clonotype clustering can be added exactly as in BLOSUM if needed. (d) *Scope*: BLOSUM is one global matrix; VDJAM is estimated per chain and per region, because (next paragraph) only the NDN core carries a transferable substitution signal.

OBSERVATION UNIT AND ESTIMATOR.. We replace “alignment columns” with *same-antigen single-substitution pairs*: ordered pairs of CDR3s annotated to the same epitope, of equal length, differing at exactly one position — the only clean substitution signal extractable without a multiple alignment. For a substitution $a \leftrightarrow b$ observed at position p in a CDR3 of length L on chain c , let $\text{bg}(\cdot | L, p)$ be the residue frequency at position p among all length- L chain- c CDR3s. With $f(a, b)$ the observed count of $\{a, b\}$ events and

$$e(a, b) = \sum_{\text{event slots } (L, p)} n_{L, p} [\text{bg}(a | L, p) \text{bg}(b | L, p) + \text{bg}(b | L, p) \text{bg}(a | L, p)], \quad \text{VDJAM}(a, b) = \log_2 \frac{f(a, b) + \alpha}{e(a, b) + \alpha}, \quad (1)$$

where $n_{L, p}$ is the number of observed events at (L, p) and α a Laplace pseudocount. The key point is that the background is *position-specific*: at an over-conserved column the residue frequency is peaked, so substitutions there are both rare (small f) and “expected” (matched e), and contribute ≈ 0 to the log-odds — the germline bias is removed analytically rather than by hand-masking. The matrix is consumed by `seqtree` via the squared-distance penalty $\text{pen}(a, b) = s_{aa} + s_{bb} - 2s_{ab}$ (`SubstitutionMatrix.from_similarity`); a dominant uniform diagonal makes higher learned sim-

ilarity rank a substitution better.

REGION STRUCTURE: NDN IS LEARNED, V/J ARE GERMLINE-DETERMINED.. A CDR₃ is V-germline prefix · NDN insert · J-germline suffix. Estimating (1) separately by region shows (Fig. 1) that the substitution signal — the agreement with general amino-acid chemistry, measured as correlation with BLOSUM62 — is concentrated in the NDN core (TRB $r=0.42$, TRA 0.35), whereas the V flank carries almost none (TRB 0.08, TRA 0.07): V-flank “substitutions” are dominated by germline allele differences, not free chemistry. This motivates a two-part model: **VDJAM is learned for the NDN core**, while the **V and J flanks are auto-computed** from the germline V/J sequences and their trimming. Concretely, a CDR₃ position derived from V- (or J-) germline is near-invariant when it survives trimming and reverts to the random-insert background when trimmed; so the V/J contribution is a per-position conservation profile $w(p) = \Pr[\text{position } p \text{ is germline-retained}]$, determined by the gene’s trimming-length distribution, rather than a substitution matrix learned from data. Combined, the NDN matrix scores the core and $w(p)$ down-weights substitutions at germline-retained flanks (realized through a per-position weighting of the base matrix). *Status*: the NDN matrix and the region decomposition are implemented and evaluated below; the germline+trimming V/J profile (from the OLGA recombination model, via mirpy) is in progress.

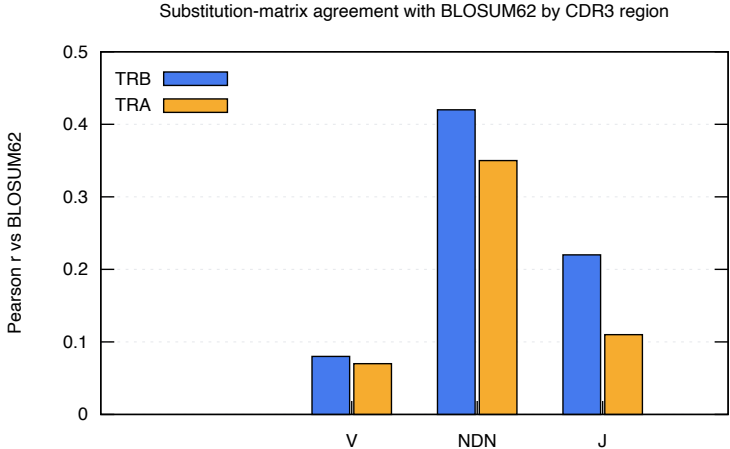


Figure 1: Per-region agreement of the learned CDR₃ substitution matrix with BLOSUM62 (Pearson r of off-diagonal scores), human VDJdb. The free-substitution signal lives in the NDN core; the V flank is germline-determined.

3. SCORE AGGREGATION AND SIGNIFICANCE

PER-HIT TO PER-QUERY.. A region-aware similarity score sums the NDN matrix score over the core and the (down-weighted) flank contributions; this is available as an optional ranking score and generalizes the legacy cloglog generalized-linear aggregation over V/CDR₁/CDR₂/J/CDR₃.

CONTROL-CALIBRATED E-VALUE (PRIMARY).. Immune repertoires are biologically redundant, so an i.i.d. null over-calls. `vdjmatch` instead counts a query’s VDJdb neighbours within the ball and com-

compares to the count expected from a matched background control: with $N = |D|$, M the control size and n_C the control neighbour count, $\mathbb{E} = (N/M) n_C$ is the expected number of VDJdb neighbours by chance and $p_{\text{enrich}} = \mathbb{P}(\text{Poisson}(\mathbb{E}) \geq n_D)$ is the enrichment significance — significant only when a clonotype has *more* VDJdb neighbours than the generative background predicts. The Poisson/Chen–Stein theory, the punctured (self-excluding) null and the choice of M are derived in the companion `seqtree` appendix.

PAIRED $\alpha\beta$ E-VALUE.. The null factorization is not an approximation of convenience: α and β chains pair *almost without constraint* in the repertoire at large, as shown by integrating structural and paired-sequencing data [1], so under the background (generative) law the joint ball mass genuinely factorizes, $\pi_0^{\alpha\beta} \approx \pi_0^\alpha \pi_0^\beta$. Crucially, the *same* analysis finds that *antigen-driven selection distorts* this uniform pairing landscape — i.e. conditioned on an epitope the two chains are *not* independent. This is exactly the structure the joint E-value exploits: the null factorizes (background pairing is unconstrained), so among the N paired VDJdb complexes the expected number of chance joint matches is $\mathbb{E}_{\alpha\beta} = N \pi_0^\alpha \pi_0^\beta$ with p the Poisson tail on the count of complexes matched in *both* chains; an excess over this factorized null is precisely the epitope-conditional dependence (co-selection of α and β) that signals a true specificity. Because the joint null is far smaller than either chain’s, true pairs are dramatically more significant: on the paired TCRvdb benchmark, 296/300 pairs match a complex and 295/296 (99.7%) recover the correct epitope, with joint p down to 3.6×10^{-9} . (When the two chains’ hits are treated as independent tests, the equivalent statistic is Fisher’s combination of the per-chain enrichments; the appendix `seqtree` co-occupancy term quantifies the residual dependence.)

4. RESULTS: DOES VDJAM HELP?

A matrix derived from same-antigen pairs must be tested on epitopes it never saw, or the evaluation is circular. For each held-out epitope E^* we re-derive the NDN matrix from *all other* epitopes, then run VDJdb-vs-VDJdb retrieval on E^* (exact self excluded), ranking neighbours by each matrix, and score the micro precision–recall AUC for “neighbour shares E^* ”. Across the eight largest TRB epitopes (scope 2, Fig. 2) the learned NDN matrix improves over unit (edit-distance) cost (mean $\Delta\text{PR-AUC} = +0.018$) and is on par with BLOSUM62 (-0.009 ; better on 4/8). At this scope the search ball already supplies most of the specific-vs-nonspecific discrimination and the matrix only re-ranks within it; the larger gains are expected from the region-aware combination (NDN matrix + germline/trimming flank weighting) and at wider scopes, which is the current line of work.

5. WORKED EXAMPLES

PAIRWISE (DIFFERENT LENGTHS).. `seqtree` aligns CDR3s of differing length, placing indels where the loop genuinely lengthens. The edit-distance alignments below (influenza GIL-specific TRB CDR3s) show the conserved CASS V-anchor and DTQYF J-motif with substitutions and indels confined to the central NDN core:

```
query (ref): CASSPLSTDTQYF    epitope GILGFVFTL    (unit / edit-distance alignment)
```

```
vs CASSRLSTDTQYF (len 13; subs=1 ins=0 dels=0; CIGAR 4=1X8=)
```

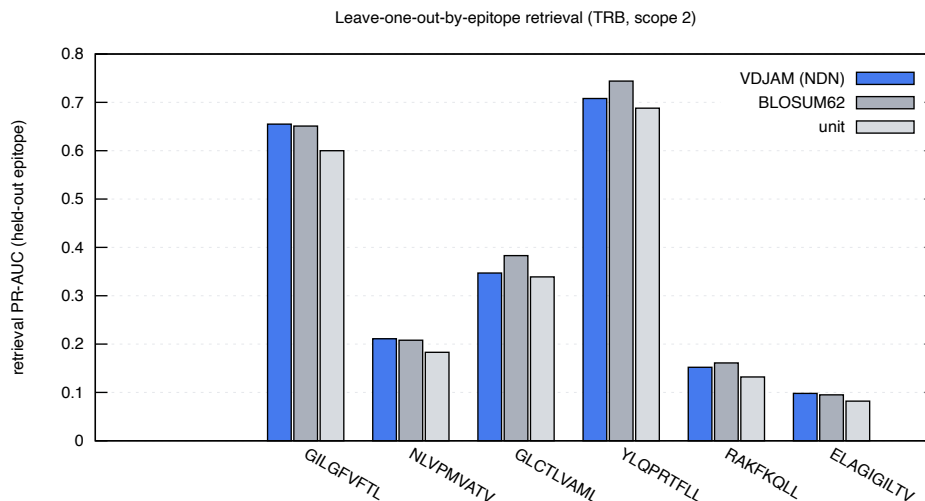


Figure 2: Leave-one-out-by-epitope retrieval PR-AUC (TRB, scope 2): the NDN VDJAM matrix versus BLOSUM62 and unit cost on epitopes excluded from matrix estimation.

```

CASSPLSTDTQYF
|||| |||||
CASSRLSTDTQYF

vs CASSHLMSTDTQYF (len 14; subs=1 ins=1 dels=0; CIGAR 4=1X1=1I7=)
CASSPL-STDTQYF
|||| | |||||
CASSHLMSTDTQYF

vs CASSFSTDTQYF (len 12; subs=1 ins=0 dels=1; CIGAR 4=1D1X7=)
CASSPLSTDTQYF
|||| |||||
CASS-FSTDTQYF

vs CASSLAPLSTDTQYF (len 15; subs=0 ins=2 dels=0; CIGAR 4=2I9=)
CASS--PLSTDTQYF
|||| |||||
CASSLAPLSTDTQYF

vs CASSGSDTQYF (len 11; subs=1 ins=0 dels=2; CIGAR 4=1D1X1=1D5=)
CASSPLSTDTQYF
|||| | ||||
CASS-GS-DTQYF

```

MULTIPLE (SAME LENGTH).. Aligning same-length same-epitope CDR₃s (no indels needed) makes the conservation profile explicit (Figs. 3–4): columns are shaded by conservation, dark at the germline-determined CASS prefix and J motif, light through the variable NDN core. This is the structural

counterpart of the finding of Shcherbinin et al. [2] that specificity-preserving CDR3 substitutions occur predominantly outside the peptide/MHC-contacting positions while preserving the loop backbone — i.e. the tolerated variation we score in the NDN core is exactly the variation that does not break recognition.

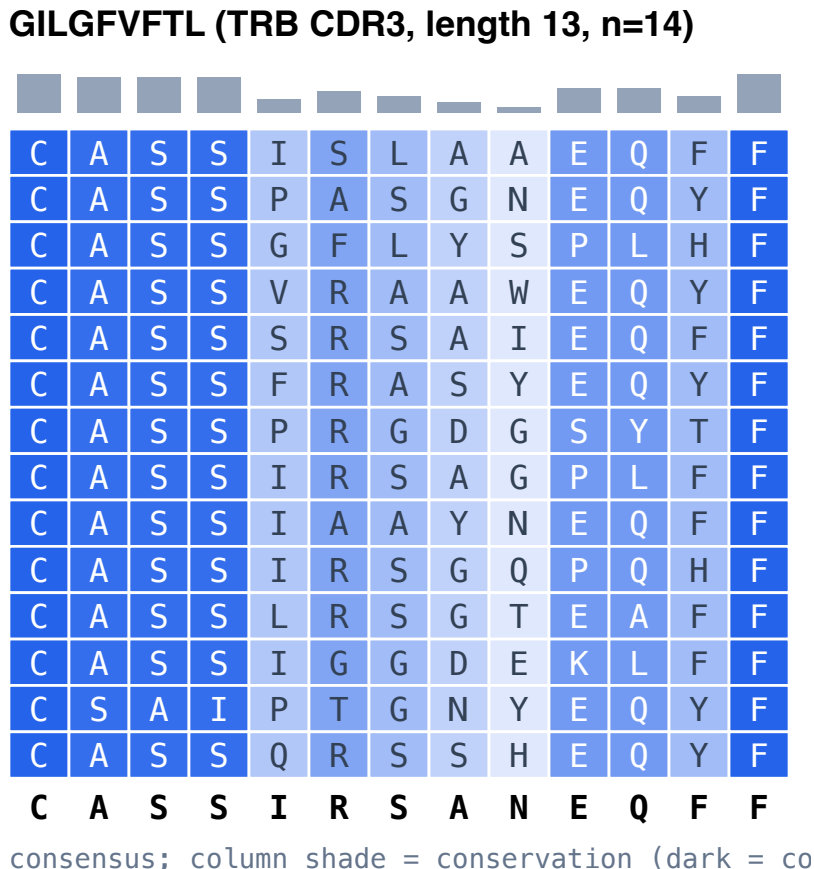


Figure 3: GIL (influenza M1, A*02) TRB CDR3 block; conserved V/J flanks, variable NDN core.

6. LIMITATIONS AND ROADMAP

The NDN matrix is estimated from a limited set of same-antigen single-substitution pairs and, at narrow scope, matches rather than beats BLOSUM62; the region-aware V/J germline+trimming model (in progress) and per-position weighting are expected to add discrimination, as is a per-chain treatment (TRA vs TRB) and a V-gene similarity term (grouping V genes by CDR1/CDR2/CDR2.5 distance). The control’s V–J usage must match the query repertoire’s, or the null is mis-specified; whether to renormalize V–J usage (at the risk of erasing genuine V-gene bias) is an open question evaluated on held-out epitopes. These extensions, and the full benchmark against existing tools, are the subject of ongoing work.

NLVPMVATV (TRB CDR3, length 15, n=14)

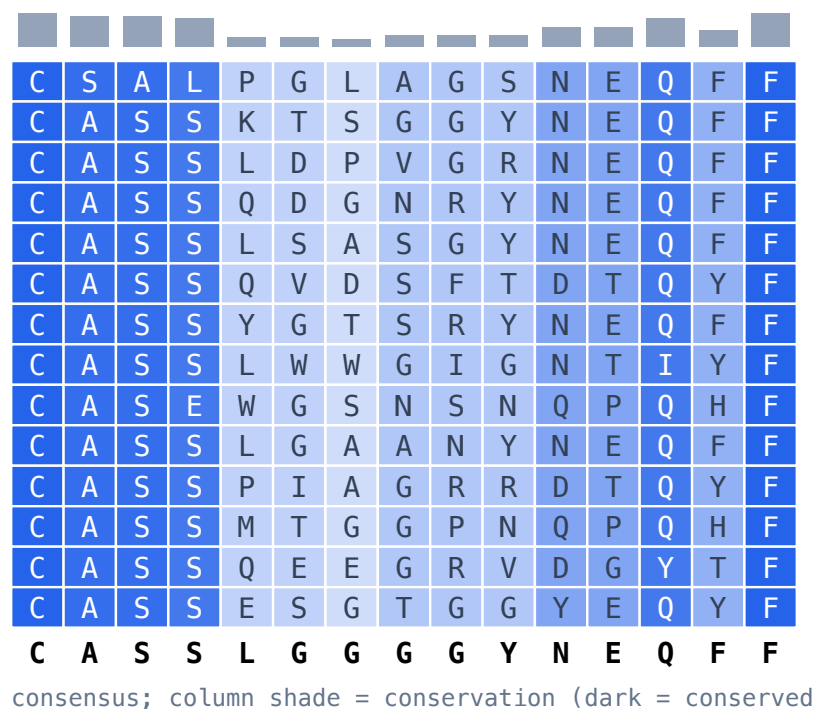


Figure 4: NLV (CMV pp65, A*02) TRB CDR3 block; same conservation pattern.

REFERENCES

- [1] Dmitrii S. Shcherbinin, Vlad A. Belousov, and Mikhail Shugay. Comprehensive analysis of structural and sequencing data reveals almost unconstrained chain pairing in TCR $\alpha\beta$ complex. *PLoS Computational Biology*, 16(3):e1007714, 2020.
- [2] Dmitrii S. Shcherbinin, Vadim K. Karnaukhov, Ivan V. Zvyagin, Dmitriy M. Chudakov, and Mikhail Shugay. Large-scale template-based structural modeling of T-cell receptors with known antigen specificity reveals complementarity features. *Frontiers in Immunology*, 14:1224969, 2023.